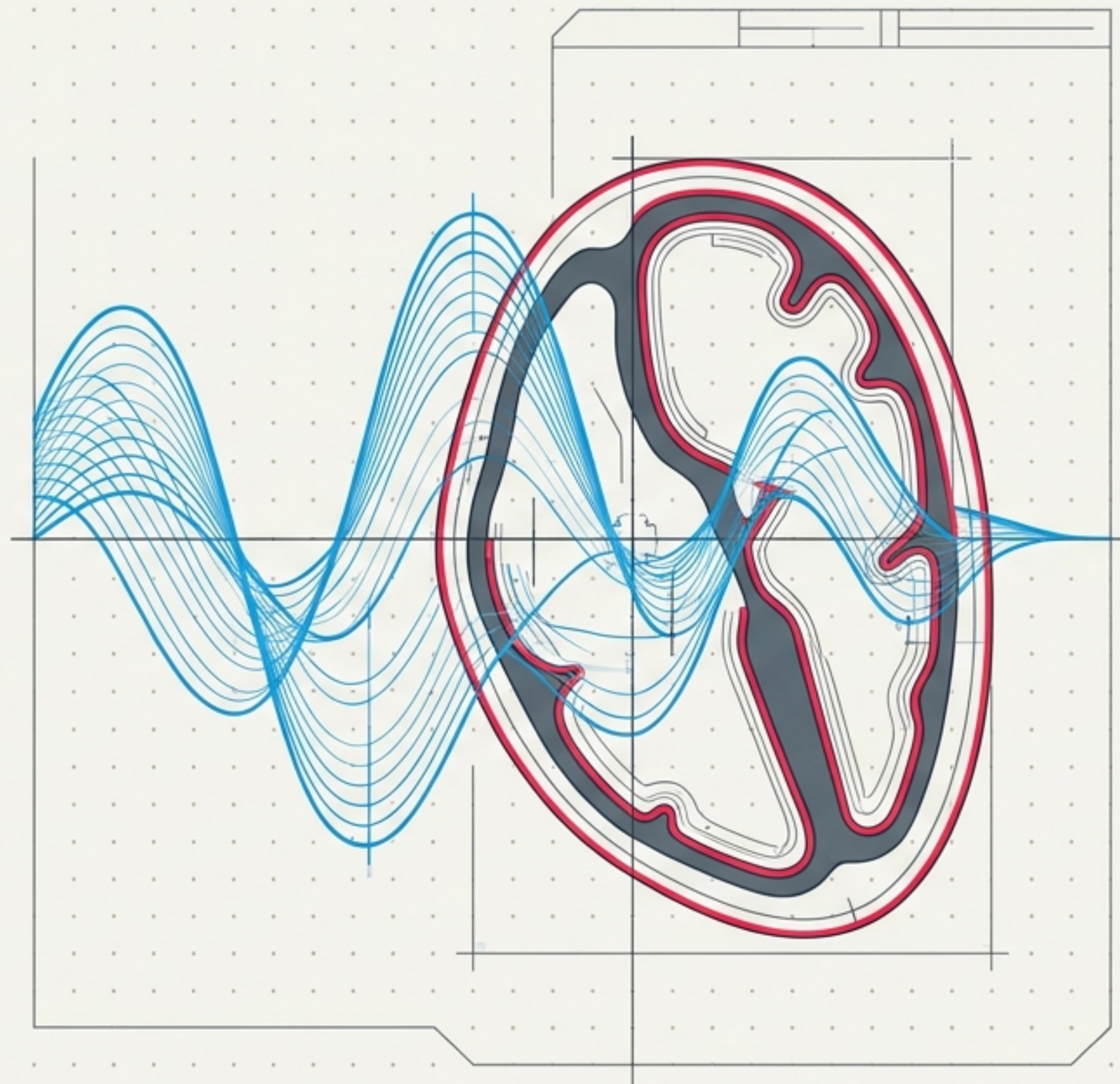


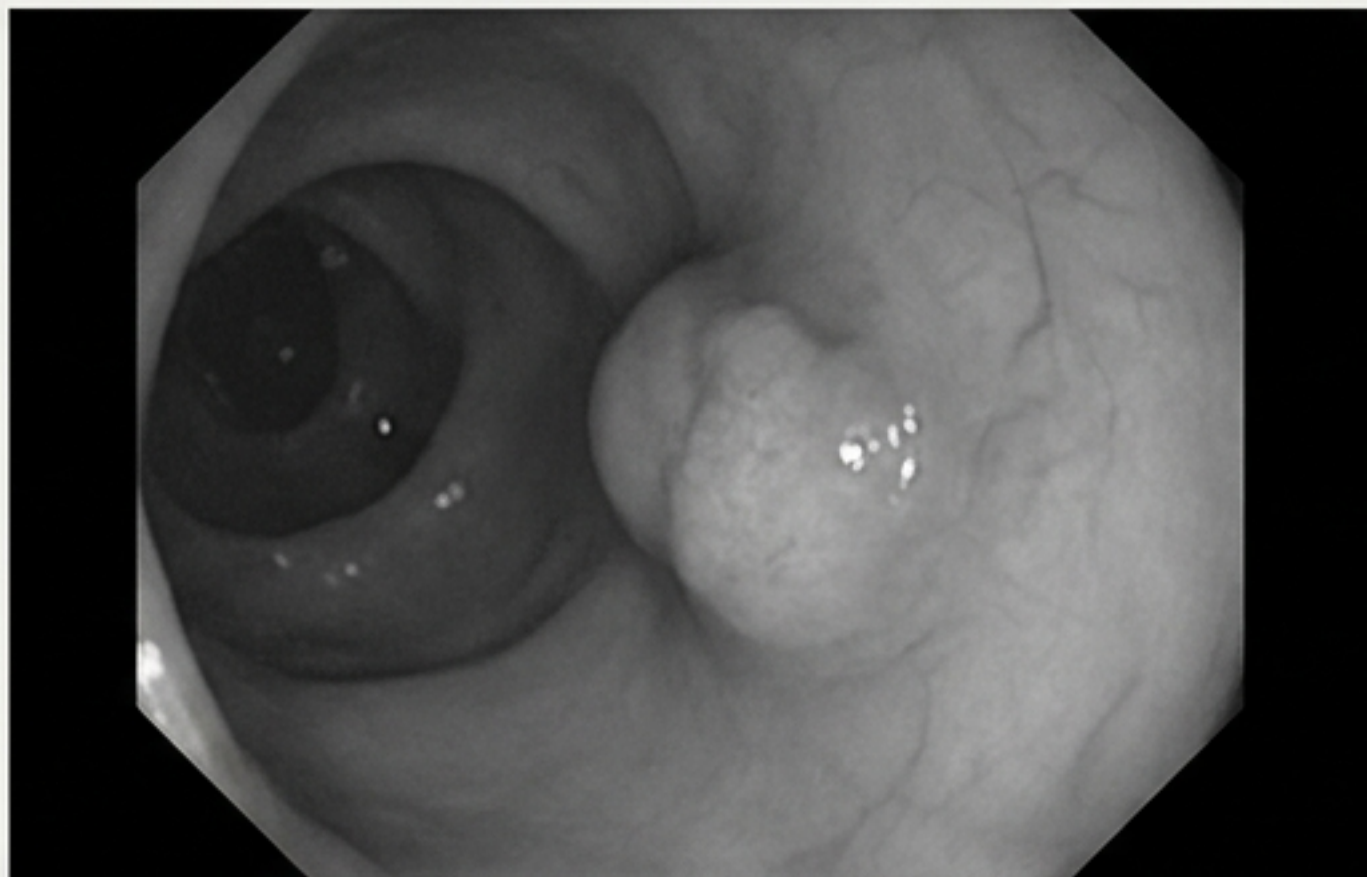
ÆMMamba: The Next Standard for 2D Medical Image Segmentation

Unifying linear complexity, global context, and absolute boundary fidelity.



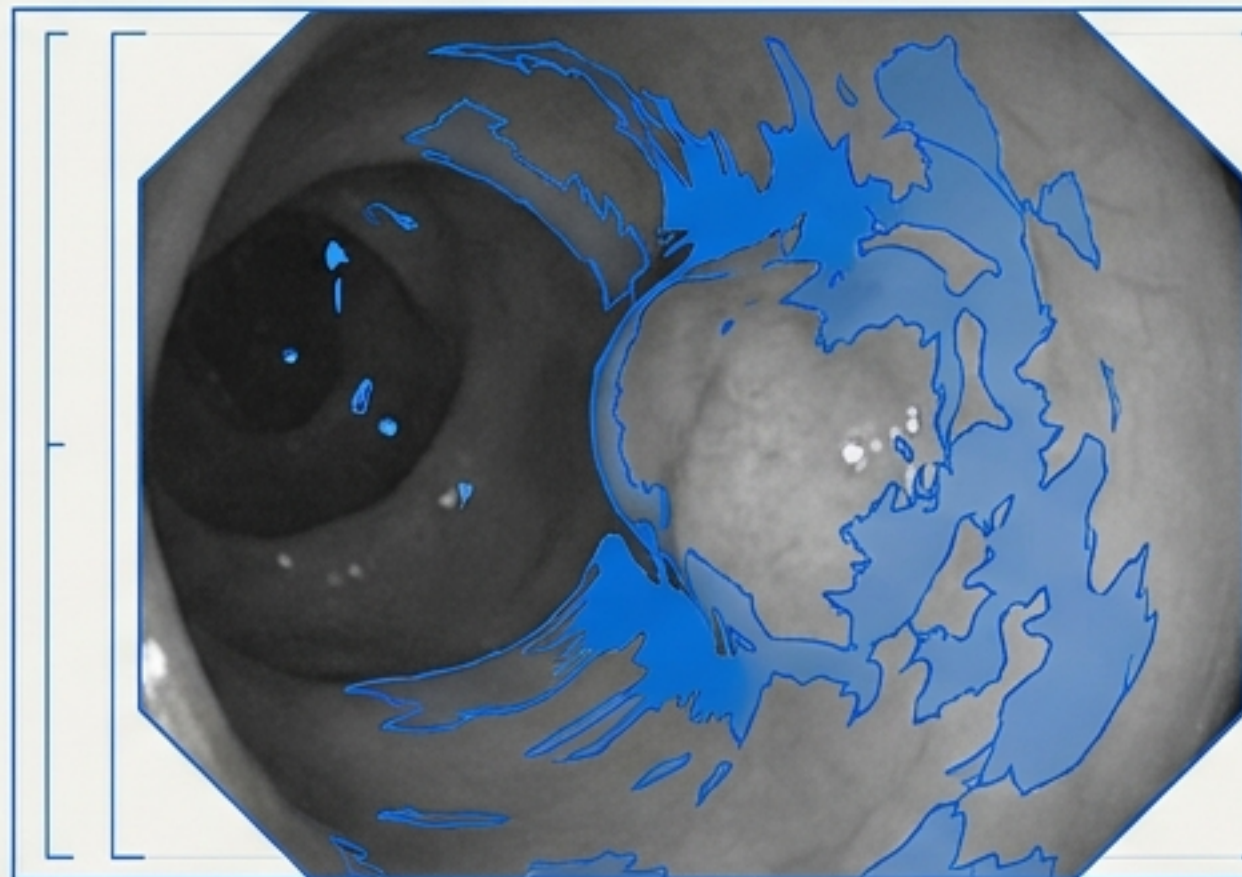
The Fuzzy Boundary Trap

The Clinical Reality



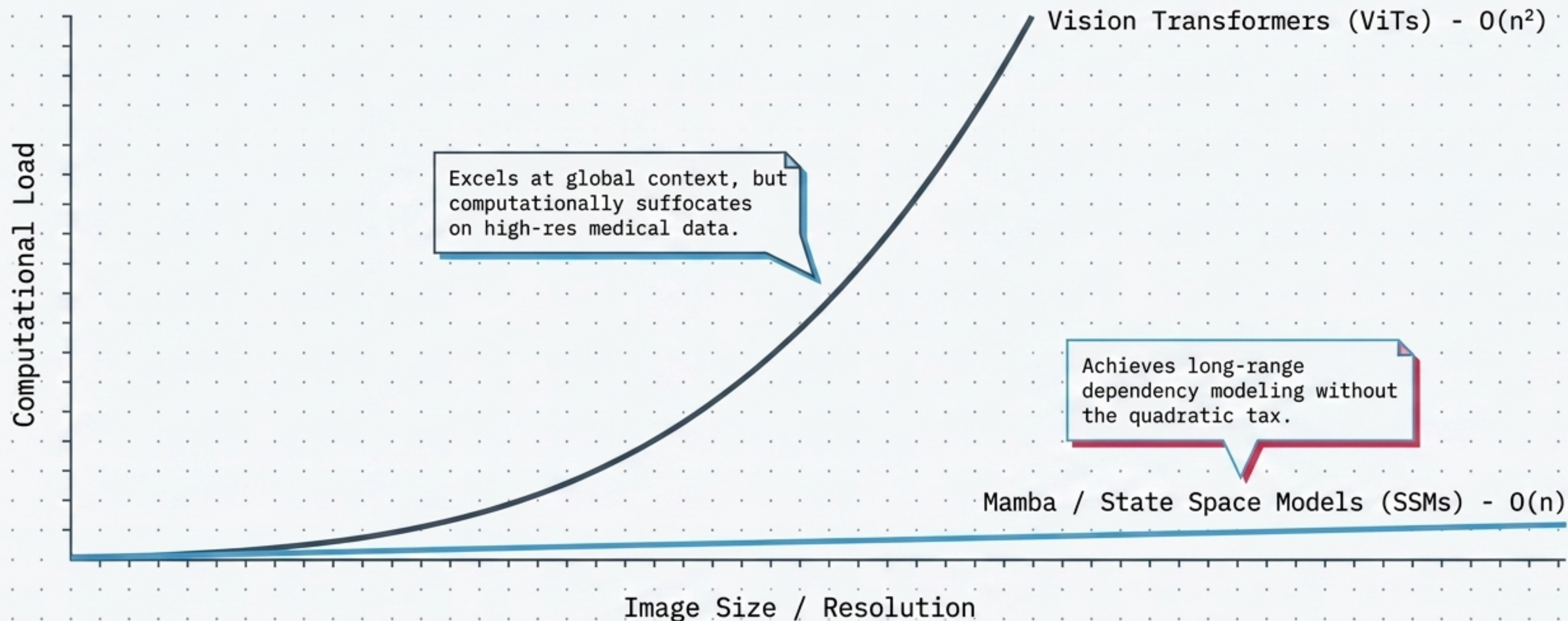
CLINICAL_CHALLENGE: High noise, low contrast, indistinct anatomical boundaries.

The Technological Failure



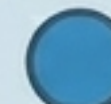
1. Legacy architectures rely heavily on distinct color contrast.
2. When pathological tissues blend, boundary delineation collapses.
3. Result: Sub-optimal disease localization and treatment planning.

The Compute vs. Context Barrier



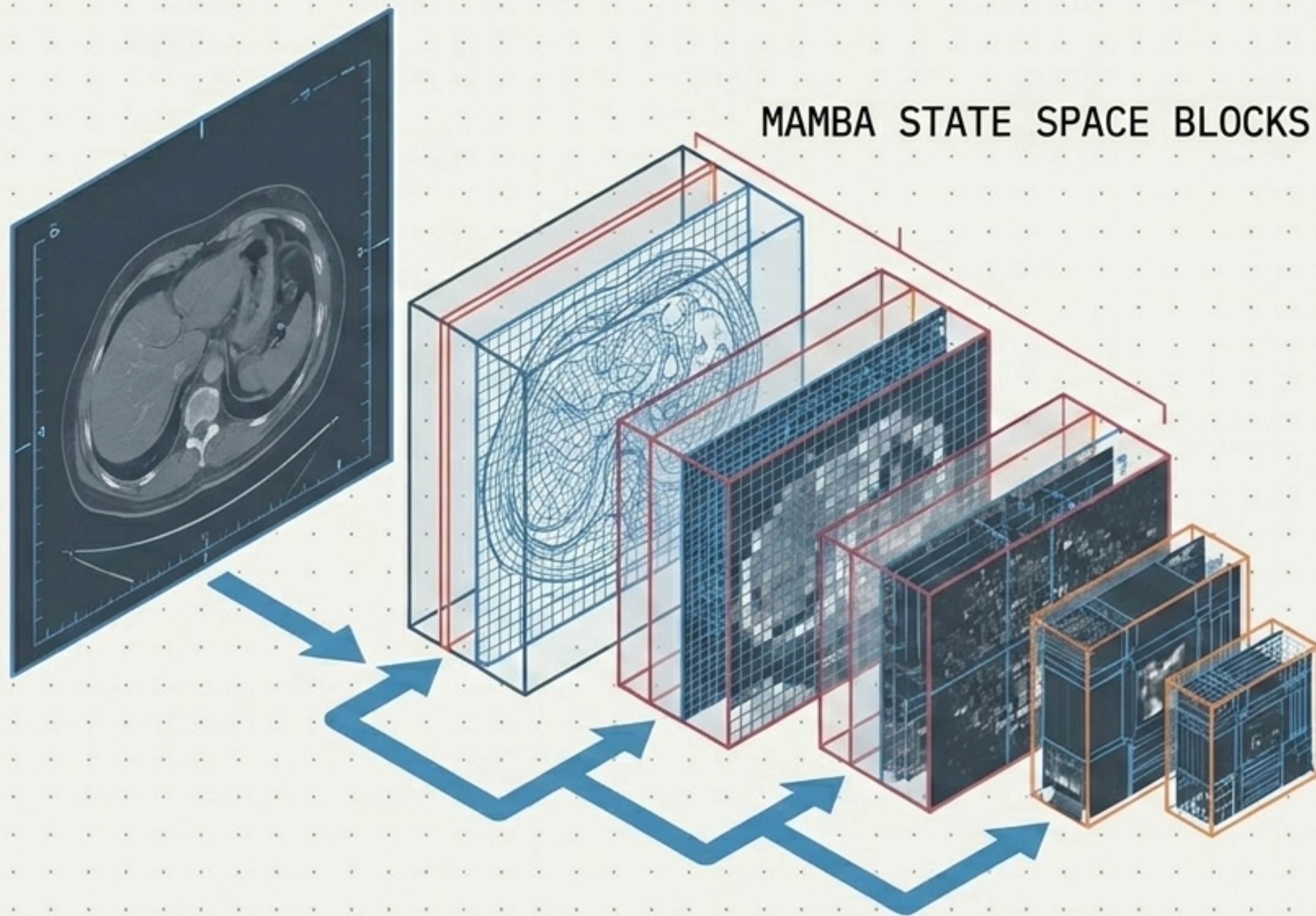
ViTs hit a scaling wall. State Space Models provide the computational breathing room required to process global anatomical context efficiently.

The Segmentation Paradigm Matrix

Features	CNNs (e.g., U-Net)	ViTs (e.g., SwinUnet)	ÆMMamba (Proposed)
Local Features			 via CNN modules
Global Context			 via SSM
Boundary Precision			 via EAM & BSD
Computational Efficiency	High - $O(n)$	Low - $O(n^2)$	High - $O(n)$
Medical Edge Cases	Fails on fuzzy bounds	Struggles on low-contrast	Bespoke Edge Design

ÆMMamba escapes the traditional trade-offs by integrating CNN-level local precision with SSM-level global efficiency.

The Engine: Multi-Scale Mamba Backbone



CORE_ARCHITECTURE:

Built upon the Mamba State Space Model.

FUNCTION:

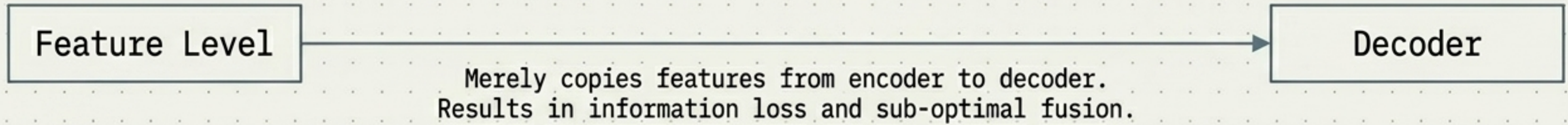
Extracts multi-scale feature representations natively.

ADVANTAGE:

Replaces heavy self-attention mechanisms with linear-complexity state spaces, spaces, processing the entire image context instantaneously.

Module 1: The Efficient Fusion Bridge (EFB)

Legacy Approach: Standard Skip Connection



EFB Approach: Active Harmonization



The EFB **actively harmonizes** multi-scale features rather than passively copying them, ensuring the decoder receives rich, complementary data.

Module 2: The Edge-Aware Module (EAM)

MECHANISM:

Sobel-based edge extraction integrated directly into the deep learning pipeline.

BEHAVIOR:

Forces the network to track subtle shifts in tissue structure rather than relying on color or brightness differences.

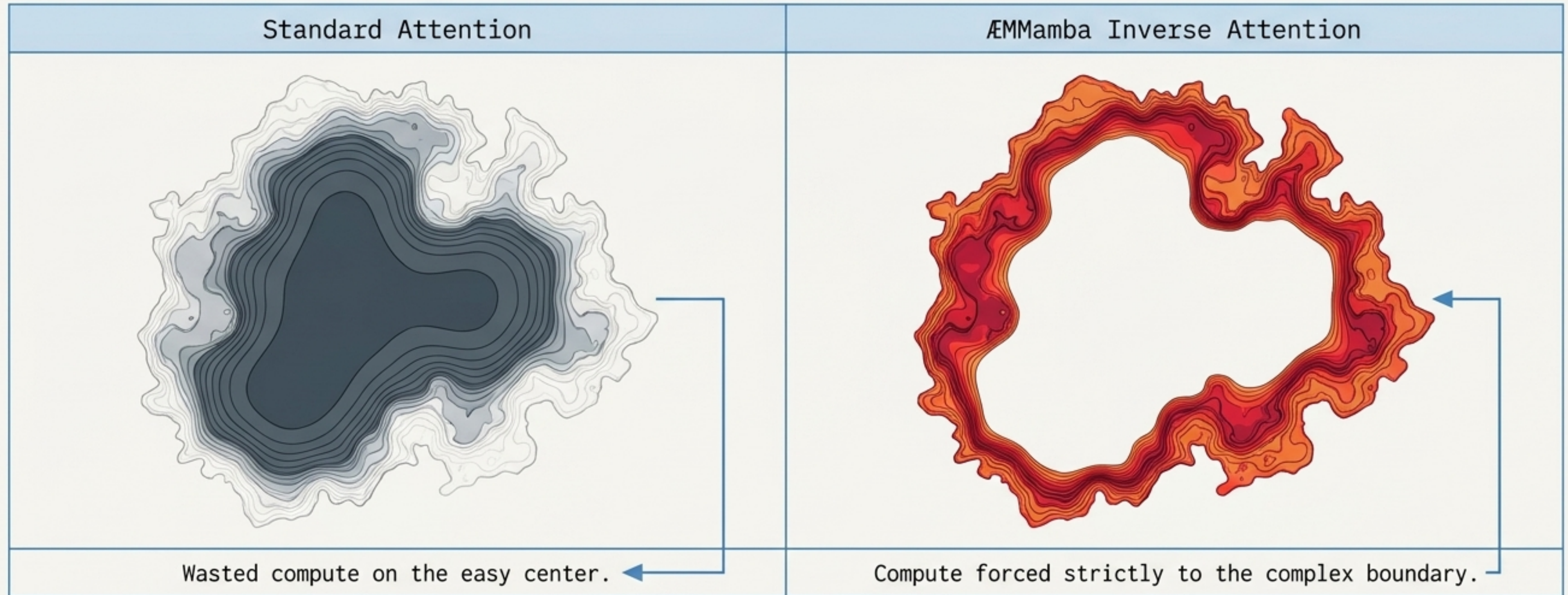


OUTCOME:

Recovers indistinct boundaries that traditional convolutional filters miss.

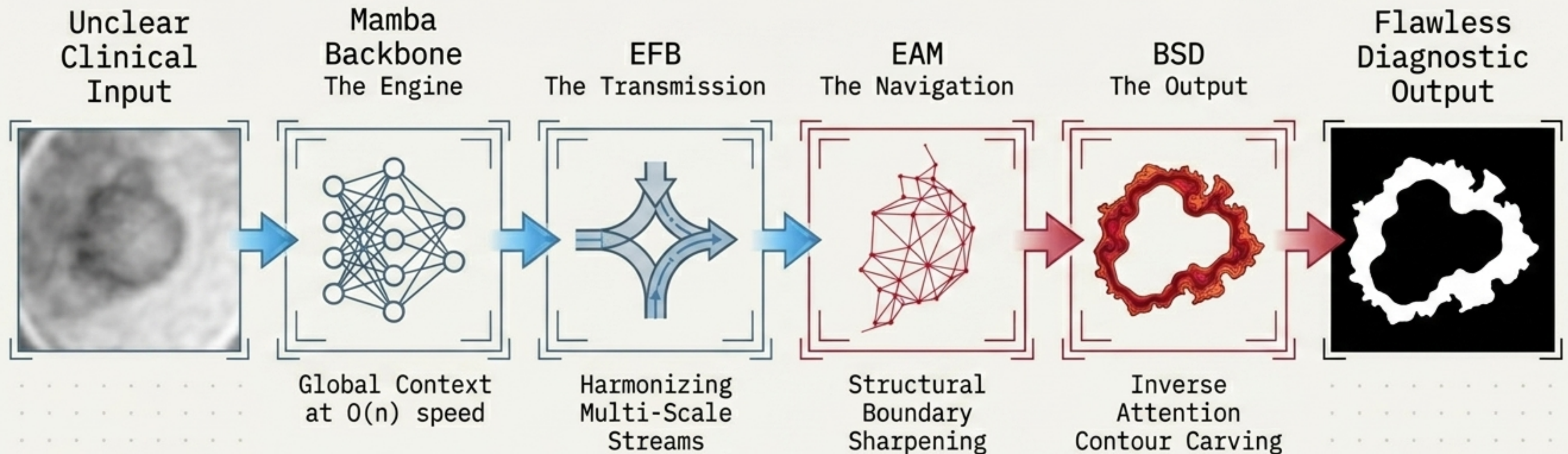
Module 3: Boundary Sensitive Decoder (BSD)

Topographical Diagnostic



The BSD utilizes Inverse Attention combined with residual convolutional layers. By actively penalizing focus on the simple core, it forces the network's processing power to sculpt cross-level complex boundaries with absolute high fidelity.

The Blueprint of Precision



ÆMMamba operates not as a disjointed collection of modules, but as a continuous pipeline engineered exclusively to preserve and enhance boundaries from input to output.

Clinical Validation: The Polyp Benchmark

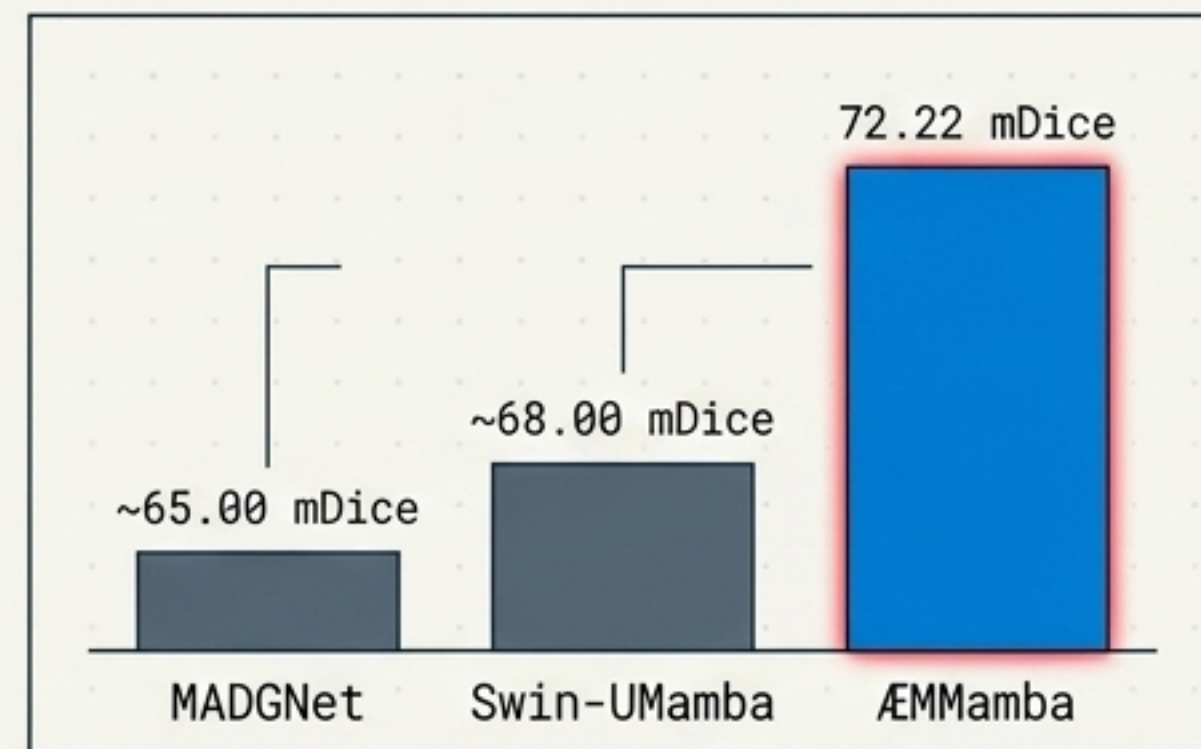
Standard Datasets

✓	Kvasir-SEG	✓	CVC-ClinicDB	✓	CVC-ColonDB	✓	CVC-EndoScene
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Achieves Highest mDice across all standard benchmarks.

The Ultimate Test - ETIS

72.22 mDice



On ETIS—widely considered the most challenging polyp dataset due to extreme low-contrast and fuzzy boundaries—AEMMamba significantly outperforms prior state-of-the-art models.

Pan-Modality Superiority

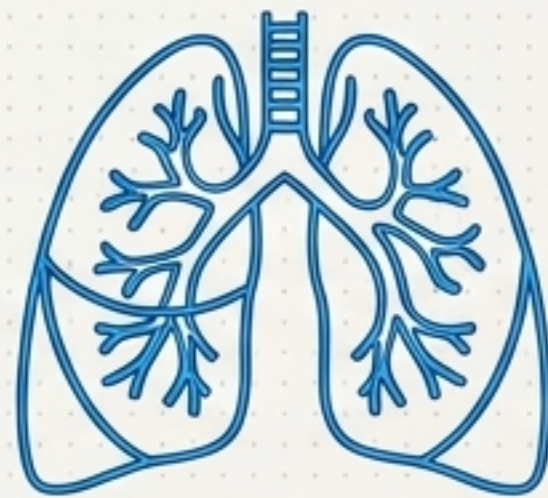


Breast Ultrasound

[DATASET: BUSI]

84.24 Dice

Surpasses H2Former & SwinUnet

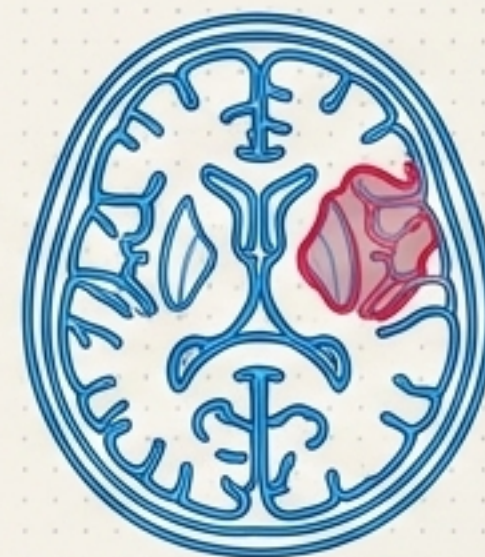


Lung Lesions

[DATASET: COVID-19 Lung]

79.83 Dice

Surpasses H2Former & SwinUnet



Brain MRI

[DATASET: LGG]

**87.25 Dice /
79.31 mIoU**

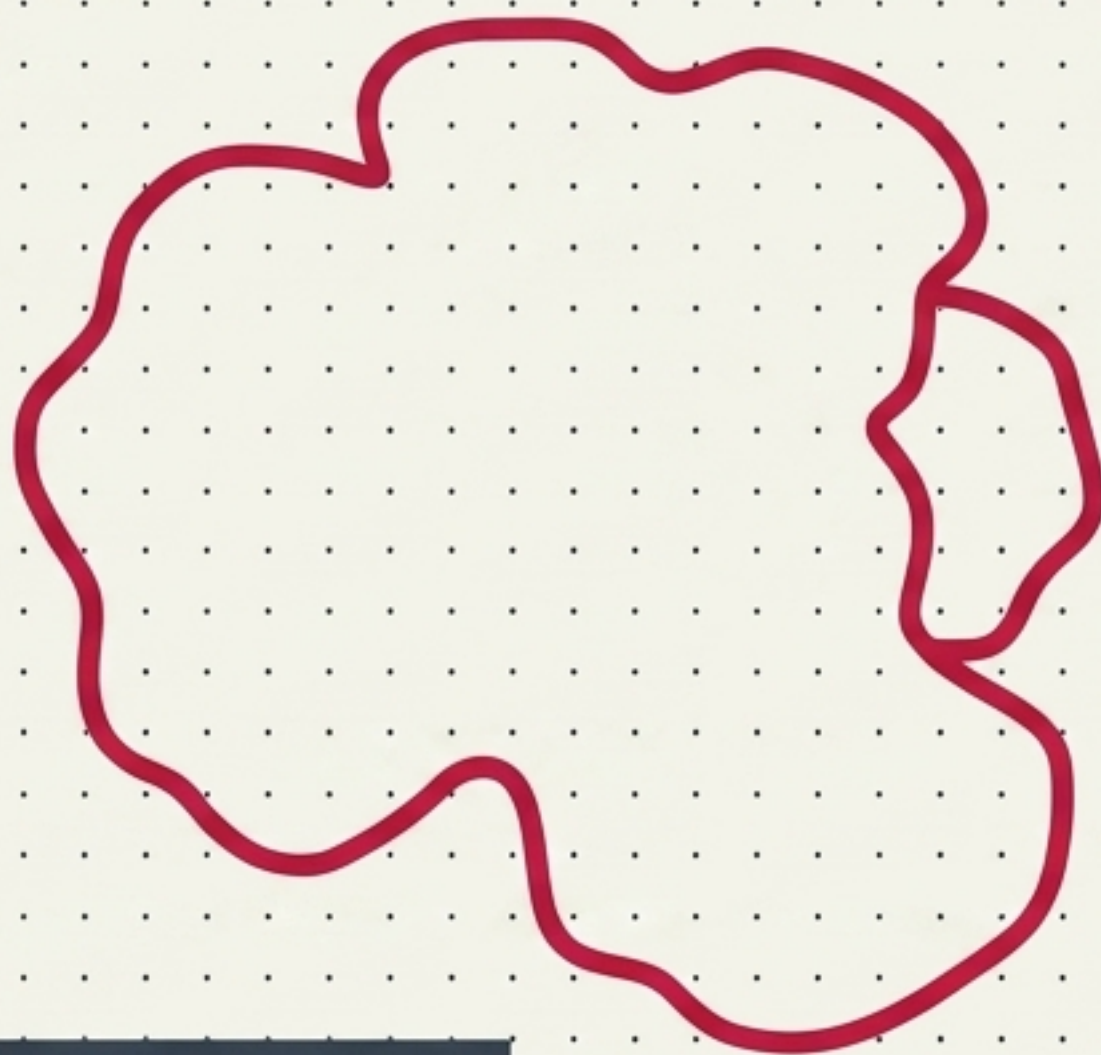
Outperforms all compared methods

Linear Efficiency

Processes global context at $O(n)$ speeds via State Space Modeling.

Absolute Fidelity

Explicitly recovers fuzzy boundaries through Sobel extraction and Inverse Attention.



Proven Superiority

Undisputed SOTA performance across 8 datasets and 4 modalities.

EMMamba successfully bridges the gap between deep technical efficiency and real-time clinical precision.